

# Displays

D1:

$$R_{[t_0, t_1]} = \{\Phi(t_1; t_0, x_0, d) : x_0 \in \mathcal{X}_0, d \in \mathcal{D}\}.$$

D2:

$$\kappa(x) = \frac{1}{z_k(x)^\top M^{-1} z_k(x)},$$

D3:

$$M = \int_{\mathbb{R}^n} z_k(x) z_k(x)^\top d\mu(x)$$

D4:

$$\kappa(x)^{-1} = z_k(x)^\top M^{-1} z_k(x),$$

D5:

$$\hat{\mu} = \frac{1}{N} \sum_{i=1}^N \delta_{x^{(i)}},$$

D6:

$$\begin{aligned} C(x) &= \hat{\kappa}^{-1}(x) = z_k(x)^\top \hat{M}^{-1} z_k(x) \\ &= z_k(x)^\top \left( \frac{1}{N} \sum_{i=1}^N z_k(x^{(i)}) z_k(x^{(i)})^\top \right)^{-1} z_k(x), \end{aligned}$$

D7:

$$N = \left\lceil \frac{5}{\epsilon} \left( \log \frac{4}{\delta} + \binom{n+2k}{n} \log \frac{40}{\epsilon} \right) \right\rceil.$$

D8:

$$\begin{aligned} \hat{M} &= \frac{1}{N} \sum_{i=1}^N z_k(x_f^{(i)}) z_k(x_f^{(i)})^\top, \\ \alpha &= \max_{i=1, \dots, N} z_k(x_f^{(i)})^\top \hat{M}^{-1} z_k(x_f^{(i)}). \end{aligned}$$

D9:

$$\hat{R}_{[t_0, t_1]} = \{x \in \mathbb{R}^n : z_k(x)^\top \hat{M}^{-1} z_k(x) \leq \alpha\}$$

D10:

$$C(x) = z_k(x)^\top \left( \frac{1}{N} \sum_{i=1}^N z_k(x^{(i)}) z_k(x^{(i)})^\top \right)^{-1} z_k(x),$$

D11:

$$N \geq \frac{5}{\epsilon} \left( \log \frac{4}{\delta} + \binom{n+2k}{n} \log \frac{40}{\epsilon} \right), \tag{1}$$

D12:

$$\begin{aligned} &\mu^N \left( \{ (x^{(1)}, \dots, x^{(N)}) : \right. \\ &\quad \left. \mu(\{x \in \mathbb{R}^n : C(x) \leq \alpha\}) \geq 1 - \epsilon \} \right) \geq 1 - \delta. \end{aligned}$$

D13:

$$\text{Pos}(V) = \{ \{x | g(x) \geq 0\}, g \in V \}$$

D14:

$$N \geq \frac{5}{\epsilon} \left( \log \frac{4}{\delta} + m \log \frac{40}{\epsilon} \right),$$

D15:

$$\begin{aligned} &\arg \min_{\alpha > 0} \quad \alpha \\ &\text{subject to} \quad z_k(x^{(i)})^\top M^{-1} z_k(x^{(i)}) \leq \alpha, \quad i = 1, \dots, N. \end{aligned}$$

D16:

$$\begin{aligned}\dot{x} &= y \\ \dot{y} &= -\alpha y + x - x^3 + \gamma \cos(\omega t),\end{aligned}\tag{2}$$

D17:

$$\alpha = 0.05, \quad \gamma = 0.4, \quad \omega = 1.3.$$

D18:

$$\begin{aligned}\ddot{x} &= u_1 K \sin(\theta) \\ \ddot{h} &= -g + u_1 K \cos(\theta) \\ \ddot{\theta} &= -d_0 \theta - d_1 \dot{\theta} + n_0 u_2,\end{aligned}$$

D19:

$$\begin{aligned}x(0) &\in [-1.7, 1.7], & \dot{x}(0) &\in [-0.8, 0.8], \\ h(0) &\in [0.3, 2.0], & \dot{h}(0) &\in [-1.0, 1.0], \\ \theta(0) &\in [-\pi/12, \pi/12], & \dot{\theta}(0) &\in [-\pi/2, \pi/2],\end{aligned}$$

D20:

$$u_1 \in [-1.5 + g/K, 1.5 + g/K], \quad u_2 \in [-\pi/4, \pi/4],$$

D21:

$$\begin{aligned}\dot{x}_1 &= \frac{1}{T} (d - \min(c, vx_1, w(\bar{x} - x_2))) \\ \dot{x}_i &= \frac{1}{T} (\min(c, vx_{i-1}, w(\bar{x} - x_i)) \\ &\quad - \min(c, vx_i, w(\bar{x} - x_{i+1}))), \quad (i = 2, \dots, n-1) \\ \dot{x}_n &= \frac{1}{T} (\min(c, vx_{n-1}, w(\bar{x} - x_n)/\beta) - \min(c, vx_n)),\end{aligned}\tag{3}$$

## Inline

I1:  $\mathbb{R}^n$

I2:  $a, b \in \mathbb{R}^n$

I3:  $[a, b] = \{x \in \mathbb{R}^n | a \leq x \leq b\}$

I4:  $\leq$

I5:  $x$

I6:  $x_i$

I7:  $i^{th}$

I8:  $x^{(i)}$

I9:  $x \in \mathbb{R}^n$

I10:  $z_k(x) \in \mathbb{R}^{\binom{n+k}{n}}$

I11:  $\leq k$

I12:  $n = 2$

I13:  $k = 2$

I14:  $z_k(x) = [1 \quad x_1 \quad x_2 \quad x_1 x_2 \quad x_1^2 \quad x_2^2]^\top$

I15:  $\leq d$

I16:  $n$

I17:  $\mathbb{R}[x]_d^n$

I18:  $z_d$

I19:  $\Phi(t_1; t_0, x_0, d)$

I20:  $x(t_0) = x_0 \in \mathbb{R}^n$

I21:  $t_0$

I22:  $t_1$

I23:  $d : [t_0, t_1] \rightarrow \mathbb{R}^w$

I24:  $\dot{x}(t) = f(t, x(t), d(t))$

I25:  $[t_0, t_1]$

I26:  $x(t_1)$

I27:  $x(t_0) = x_0$

I28:  $d$   
 I29:  $\mathcal{X}_0 \subset \mathbb{R}^n$   
 I30:  $\mathcal{D}$   
 I31:  $\mathcal{X}_0$   
 I32:  $X_0$   
 I33:  $D$   
 I34:  $\Phi(t_1; t_0, X_0, D)$   
 I35:  $\mu$   
 I36:  $\mu(A)$   
 I37:  $A \in \mathbb{R}^n$   
 I38:  $x_0$   
 I39:  $\Phi(t_1; t_0, x_0, d)$   
 I40:  $A$   
 I41:  $A \subseteq \mathbb{R}^n$   
 I42:  $B \subseteq \mathbb{R}^n$   
 I43:  $B$   
 I44:  $\hat{R}_{[t_0, t_1]}$   
 I45:  $\mu(\hat{R}_{[t_0, t_1]})$   
 I46:  $\Phi(t_1; t_0, x_0, u)$   
 I47:  $\epsilon \in (0, 1)$   
 I48:  $\mu(\hat{R}_{[t_0, t_1]}) \geq 1 - \epsilon$   
 I49:  $\hat{R}_{[t_0, t_1]} = \mathbb{R}^n$   
 I50:  $\mu(\hat{R}_{[t_0, t_1]}) = 1$   
 I51:  $k$   
 I52:  $M$   
 I53:  $z_k(x)$   
 I54:  $M^{-1}$   
 I55:  $2k$   
 I56:  $i = 1, \dots, N$   
 I57:  $\delta_x$   
 I58:  $\int f(y) d\delta_x(y) = f(x)$   
 I59:  $\hat{\mu}$   
 I60:  $\hat{M}$   
 I61:  $\hat{M}^{-1}$   
 I62:  $N \geq \binom{n+k}{n}$   
 I63:  $C(x)$   
 I64:  $\alpha \in \mathbb{R}$   
 I65:  $\{x \in \mathbb{R}^n : C(x) \leq \alpha\}$   
 I66:  $\Phi$   
 I67:  $[t_0, t_1]$   
 I68:  $\epsilon$   
 I69:  $\delta$   
 I70:  $1 - \delta$   
 I71:  $i \in \{1, \dots, N\}$   
 I72:  $x_0^{(i)}$   
 I73:  $d^{(i)}$   
 I74:  $x_f^{(i)} = \Phi(t_1; t_0, x_0^{(i)}, d^{(i)})$   
 I75:  $\alpha$   
 I76:  $C$   
 I77:  $x^{(1)}, \dots, x^{(N)}$   
 I78:  $\alpha = \max_i C(x^{(i)})$   
 I79:  $\mu^N$   
 I80:  $N$   
 I81:  $\geq 1 - \delta$   
 I82:  $1 - \epsilon$   
 I83:  $\delta = 10^{-9}$

I84:  $V$   
 I85:  $g : \mathbb{R}^n \rightarrow \mathbb{R}$   
 I86:  $m$   
 I87:  $\mathcal{C}$   
 I88:  $c \in \mathcal{C}$   
 I89:  $\hat{\ell}(c) = \frac{1}{N} \sum_{i=1}^N 1\{x^{(i)} \notin c\}$   
 I90:  $\ell(c) = \mathbb{E}_\mu[1\{X \notin c\}] = 1 - \mu(c)$   
 I91:  $\hat{\ell}(c) = 0$   
 I92:  $x^{(i)}, i = 1, \dots, n$   
 I93:  $c$   
 I94:  $\mu^N(\{x^{(1)}, \dots, x^{(N)} : \ell(c) \leq \epsilon\}) \geq 1 - \delta$   
 I95:  $c = \{x \in \mathbb{R}^n | C(x) \leq \alpha\}$   
 I96:  $\mathcal{C} = \text{Pos}(\mathbb{R}[x]_d^n)$   
 I97:  $\binom{n+2k}{n}$   
 I98:  $z(x)^\top \hat{M}^{-1} z(x)$   
 I99:  $(x_1, \dots, x_n) \in \mathbb{R}^n$   
 I100:  $x_1, \dots, x_m$   
 I101:  $m < n$   
 I102:  $x_f^{(i)}$   
 I103:  $\mathbb{R}^m$   
 I104:  $x, y \in \mathbb{R}$   
 I105:  $\alpha, \gamma, \omega \in \mathbb{R}$   
 I106:  $\gamma$   
 I107:  $\omega$   
 I108:  $x(0) \in [0.95, 1.05]$   
 I109:  $y(0) \in [-0.05, 0.05]$   
 I110:  $[t_0, t_1] = [0, 100]$   
 I111:  $k = 10$   
 I112:  $\epsilon = 0.05$   
 I113:  $N = 156, 626$   
 I114:  $156, 626$   
 I115:  $N_{ap}$   
 I116:  $N_{out}$   
 I117:  $1 - N_{out}/N_{AP}$   
 I118:  $N_{AP} = 46, 052$   
 I119:  $1 - (2 \times 10^{-5})$   
 I120:  $0.99 - 2 \times 10^{-5}$   
 I121:  $0.95$   
 I122:  $h$   
 I123:  $\theta$   
 I124:  $\theta = 0$   
 I125:  $u_1$   
 I126:  $u_2$   
 I127:  $g = 9.81$   
 I128:  $K = 0.89/1.4$   
 I129:  $d_0 = 70$   
 I130:  $d_1 = 17$   
 I131:  $n_0 = 55$   
 I132:  $u_1(t) = u_1$   
 I133:  $u_2(t) = u_2$   
 I134:  $\forall t \in [t_0, t_1]$   
 I135:  $[t_0, t_1] = [0, 5]$   
 I136:  $(x, h)$   
 I137:  $k = 4$   
 I138:  $n = 6$   
 I139:  $N = 2, 009, 600$

I140:  $N = 32, 292$   
 I141:  $T$   
 I142:  $x_1, \dots, x_n$   
 I143:  $x_1$   
 I144:  $v$   
 I145:  $\bar{x}$   
 I146:  $w$   
 I147:  $u$   
 I148:  $T = 30$   
 I149:  $v = 0.5$   
 I150:  $w = 1/6$   
 I151:  $\bar{x} = 320$   
 I152:  $x_i(0) \in [100, 200]$   
 I153:  $i = 1, \dots, n$   
 I154:  $d \in [40/T, 60/T]$   
 I155:  $[t_0, t_1] = [0, 4T]$   
 I156:  $x^{(1)}(0)$   
 I157:  $x^{(2)}(0)$   
 I158:  $d^{(1)}, d^{(2)}$   
 I159:  $x^{(1)}(0) \leq x^{(2)}(0)$   
 I160:  $d^{(1)}(t) \leq d^{(2)}(t), t \in [0, T]$   
 I161:  $x^{(1)}(T) \leq x^{(2)}(T)$   
 I162:  $\underline{x}$   
 I163:  $\bar{x}$   
 I164:  $\underline{d}$   
 I165:  $\bar{d}$   
 I166:  $[\Phi(t_1; t_0, \underline{x}, \underline{d}), \Phi(t_1; t_0, \bar{x}, \bar{d})]$   
 I167:  $x_5$   
 I168:  $x_6$   
 I169:  $\phi_k(x)$   
 I170:  $(1 + x^\top x)^k$   
 I171:  $\phi(x)^\top \phi(x) = (1 + x^\top x)^k$   
 I172:  $\text{Pos}(\mathbb{R}[x]_d^n)$   
 I173:  $z(x)$   
 I174:  $\Phi(t_1; t_0, x_0, u)$   
 I175:  $N_{AP}$   
 I176:  $\beta$